Halward Inspection Services Ltd.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
HIS-2010-0318-DS-2-RV-102	Issued	2010-09-13

UT Thickness Data Sheet - 2-RV-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 2-RV-102, located within the ATU-100. The scope of work is limited to the thickness survey of this specific regenerator overhead line as part of the activities detailed in HIS-2010-0318.

Kevin Doucette, acting as the lead UT technician and SNT-TC-1A UT Level II for Halward Inspection Services Ltd., performed the measurements and is responsible for the accuracy of the recorded data.

Technical Specifications

The subject line, 2-RV-102, is constructed from A106 Gr.B with an outer diameter of 60.3 mm and a nominal wall thickness of 5.54 mm. The design parameters for this circuit are 345 kPa(g) and 121 °C.

The minimum allowable thickness (t-min) for this line is 2.79 mm. Any reading recorded below this value is subject to an NCR per site retirement thickness rules.

Procedural Compliance

Thickness measurements were performed under the requirements of QSCP-NDT-001 Rev 1, which mandates the use of gauges with waveform display and requires scanning at elbows and tees rather than spot measurements.

The inspection was also conducted in accordance with HIS-UT-002 Rev 1, the company procedure submitted for QSCP acceptance, and the guidelines set forth in ASTM E797.

Inspection Interval and Analysis

The regenerator overhead circuit is assigned a 3-year thickness interval. This survey supports the 3-year regenerator overhead survey and the baseline of the new 4-RA-104 spool as cited in HIS-2010-0318.

Remaining life and the next inspection date are calculated based on the larger of the long-term or short-term corrosion rates, with the next reading due at no more than half the remaining life and within the Class 1 maximum of 5 years.

Thickness readings, measured 2010-09-13

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2007-09-18	6.95	6.78	0.056	0.044	>50	2013-09-13	
3-RV-101-02	7.62	3.56	2007-09-18	6.89	6.59	0.068	0.075	40.6	2013-09-13	
3-RV-101-03	7.62	3.56	2007-09-18	6.56	6.16	0.094	0.090	27.8	2013-09-13	
3-RV-101-04	7.62	3.56	2007-09-18	6.66	6.43	0.080	0.048	35.9	2013-09-13	
3-RV-101-05	7.62	3.56	2007-09-18	6.87	6.63	0.072	0.084	36.8	2013-09-13	
3-RV-101-06	7.62	3.56	2007-09-18	6.71	6.43	0.080	0.068	35.9	2013-09-13	
3-RV-101-07	7.62	3.56	—	—	6.20	—	—	—	2013-09-13	
3-RV-101-08	7.62	3.56	—	—	6.71	—	—	—	2013-09-13	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2007-09-18	4.83	4.55	0.068	0.066	26.1	2013-09-13	
2-RV-102-02	5.54	2.79	2007-09-18	4.67	4.60	0.068	0.053	26.5	2013-09-13	
2-RV-102-03	5.54	2.79	2007-09-18	5.01	4.84	0.044	0.066	31.1	2013-09-13	
2-RV-102-04	5.54	2.79	2007-09-18	5.12	4.84	0.051	0.051	40.2	2013-09-13	
2-RV-102-05	5.54	2.79	—	—	5.04	—	—	—	2013-09-13	
2-RV-102-06	5.54	2.79	—	—	4.82	—	—	—	2013-09-13	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-03	8.56	3.81	—	—	8.59	—	—	—	2020-09-13	
4-RA-104-04	8.56	3.81	—	—	8.52	—	—	—	2020-09-13	
4-RA-104-05	8.56	3.81	—	—	8.43	—	—	—	2020-09-13	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2010-09-13	_____